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**DESIGN OF GRAPHITE NOSETIPS FOR BALLISTIC REENTRY**

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# **AIAA 5th Fluid and Plasma Dynamics Conference**

**BOSTON, MASSACHUSETTS / JUNE 26-28, 1972**

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# DESIGN OF GRAPHITE NOSETIPS FOR BALLISTIC REENTRY

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## Abstract

Analyses are given for shape change and indepth response of contemporary low-recession graphite nosetips. Both blunt and sharp shell designs and blunt plug configurations are treated from the principal viewpoint of surviving thermally induced tensile strains. Parametric calculations demonstrate the magnitude and variation in these reentry thermal strains and their sensitivity to initial reentry conditions and to various thermal and mechanical modeling elements in the design analysis sequence. In addition to thermal strain, a number of other design problems peculiar to graphite nosetips are examined in their relationship to overall reentry vehicle thermal/structural capability.

## Nomenclature

|               |                                                                              |
|---------------|------------------------------------------------------------------------------|
| $B_w$         | mass-transfer parameter                                                      |
| $c$           | specific heat (Btu/lb-°F)                                                    |
| $d$           | nosetip overhang radius (ft)                                                 |
| $E$           | elastic modulus (lb/in <sup>2</sup> )                                        |
| $g$           | acceleration of gravity (ft/sec <sup>2</sup> )                               |
| $H$           | enthalpy (Btu/lb)                                                            |
| $h$           | heat-transfer coefficient (lb/sec-ft <sup>2</sup> )                          |
| $k$           | thermal conductivity (Btu/sec-ft-°F)                                         |
| $Ma$          | Mach number                                                                  |
| $m$           | mass (lb)                                                                    |
| $p$           | pressure (atm)                                                               |
| $Q$           | total convective heat load (Btu/ft <sup>2</sup> )                            |
| $q$           | instantaneous heat rate (Btu/sec-ft <sup>2</sup> )                           |
| $\bar{q}$     | dynamic pressure (atm)                                                       |
| $R$           | radius of curvature (ft)                                                     |
| $Re$          | Reynolds number                                                              |
| $R_N$         | nosetip radius (ft)                                                          |
| $r$           | radial distance from centerline (ft)                                         |
| $S$           | surface distance from stagnation point (ft)                                  |
| $\dot{s}$     | surface recession rate (ft/sec)                                              |
| $s$           | total stagnation-point recession (ft)                                        |
| $t$           | temperature (°F)                                                             |
| $V_E$         | reentry velocity (ft/sec)                                                    |
| $x$           | axial distance from stagnation point (ft)                                    |
| $y$           | normal distance from original heatshield surface (ft)                        |
| $z$           | altitude (ft)                                                                |
| $\beta$       | thermal-expansion coefficient (in/in-°F)                                     |
| $\bar{\beta}$ | ballistic coefficient (lb/ft <sup>2</sup> )                                  |
| $\Gamma$      | permeability (ft <sup>2</sup> )                                              |
| $\gamma_E$    | reentry angle (°)                                                            |
| $\delta$      | nosetip overhang length (ft), deflection (ft)                                |
| $\epsilon$    | thermal strain (%)                                                           |
| $\zeta$       | mechanical erosion amplification factor = $\dot{s}_{total}/\dot{s}_{chem}$   |
| $\theta$      | time (sec)                                                                   |
| $\lambda$     | reentry parameter = $\sigma(V_E^2/g)\sqrt{\beta/\rho_0 R_N^3 \sin \gamma_E}$ |
| $\nu$         | Poisson ratio                                                                |
| $\rho$        | air density (lb/ft <sup>3</sup> )                                            |
| $\Gamma$      | constant                                                                     |

## Subscripts:

|        |                                  |
|--------|----------------------------------|
| $c$    | cavity, critical (onset) value   |
| $chem$ | chemical                         |
| $E$    | reentry                          |
| $e$    | boundary-layer edge value        |
| $hoop$ | hoop (circumferential) direction |

|          |                                         |
|----------|-----------------------------------------|
| $N$      | nosetip                                 |
| $r$      | radial                                  |
| $r-x$    | pitch plane                             |
| $sp$     | stagnation point                        |
| $t$      | tensile                                 |
| $x$      | axial                                   |
| $w$      | wall (surface or internal) value, wedge |
| $0$      | sea level                               |
| $\theta$ | momentum thickness                      |

## Introduction

An increasing number of unclassified technology papers have appeared in recent years on the subject of ballistic-reentry nosetips. Some have considered reinforced plastics [1] and others reinforced carbon nosetip materials [2,3]. Papers dealing with graphite nosetips have concentrated primarily on shape change and thermal response [4,5,6], and with ground testing of materials and subscale nosetip configurations [7,8,9]. No complete papers appear to treat the combined thermal/structural design problems of contemporary graphite nosetips.

The present paper deals with axisymmetric graphite nosetips at zero angle of attack. The technology base for an integrated thermal/structural design methodology is described and typical response and sensitivity results are given for two specific classes of graphite nosetips, namely sharp and blunt shell designs and blunt plug configurations.

## Design Problem

Figure 1 illustrates some general configurational features of ballistic-reentry nosetips, and

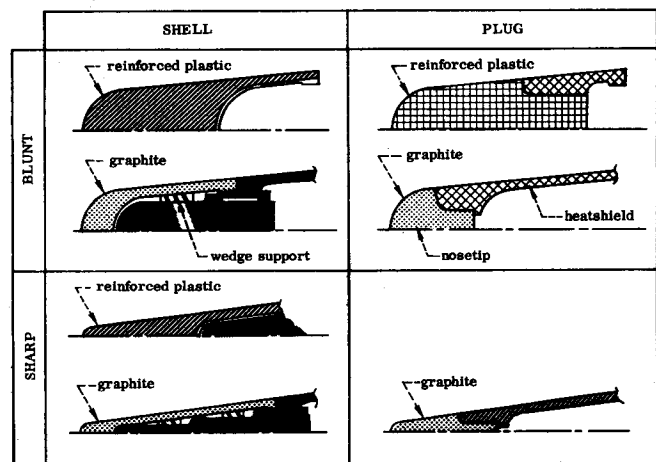


FIG. 1. TYPICAL BALLISTIC REENTRY NOSETIPS.

Fig. 2 gives a number of possible reentry failure modes for the particular graphite shell and plug nosetips of interest here. Failures can result from shape-change effects, thermal-expansion mismatches, or fracture of the brittle graphite caused by excessive thermal strains. Since the shell nosetip is

Numbers in brackets refer to references.

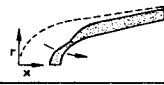
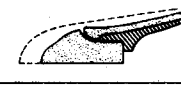
|                   | SHELL                                                                                                                                                                                                                | PLUG                                                                                                                                                                                                                   |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SHAPE CHANGE      | <br>ablation burnthrough                                                                                                            | <br>aggravated heatshield ablation                                                                                                    |
| THERMAL EXPANSION | <br>shell/wedge growth mismatch                                                                                                     | <br>plug/heatshield growth mismatch                                                                                                   |
| THERMAL STRAIN    | <br>excessive r-x strain <br>excessive hoop strain | <br>excessive axial strain <br>excessive hoop strain |

FIG. 2. EXAMPLE NOSETIP FAILURE MODES.

generally designed for minimum wall thickness, burn-through of the forward shell or sidewall can result from excessive surface recession (ablation uncertainties, perturbed trajectory, and angle of attack). In addition, a thin shell can allow external boundary-layer gases to leak into and through the shell owing to the open porosity of crystalline graphite. Shape change of a plug nosetip effects downstream heat-shield ablation. Also, a plug nosetip can lose lateral support and shape stability near the plug/heatshield interface due to indepth and backface thermal decomposition of the surrounding heatshield which is used to hold and support the high-temperature graphite plug. Thermal-expansion effects are strongly dependent on details of the support structure and attachment schemes. Local overloading and/or lateral shifting of a shell nosetip can result from mismatches in radial growth between segmented wedge supports and the surrounding graphite shell.

Generally the most critical mode of failure for graphite nosetips is fracture caused by excessive thermal strains that develop during reentry. Fracture can initiate along the inner surface of a shell nosetip or along the axis in the thicker portion of a plug configuration (Fig. 2). The pattern of these tensile cracks may be longitudinal or circumferential depending on direction of maximum strain and on material capability. Figure 3 suggests that peak tensile thermal strains in a graphite nosetip are generally

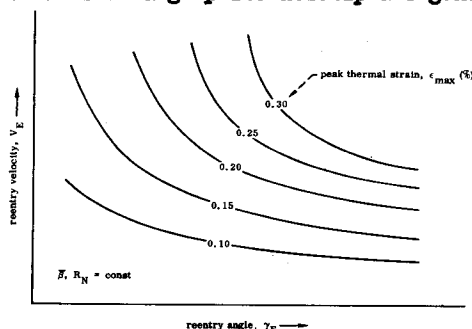


FIG. 3. SCHEMATIC V- $\delta$  ISOSTRESS MAP.

higher as the initial reentry velocity and/or angle increases. However, larger  $V_E$  and/or  $\gamma_E$  increases surface pressure and recession, both of which tend to relieve thermal strains. As such, maximum thermal strains do not necessarily occur in the upper-right regions of the V- $\delta$  map.

## Design Approach

Figure 4 gives a simplified flow diagram for the nosetip design analysis sequence used here.

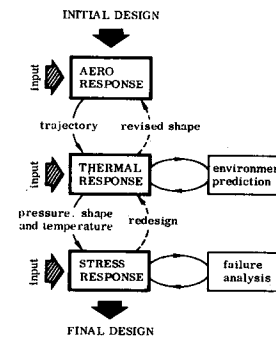


FIG. 4. NOSETIP DESIGN ANALYSIS SEQUENCE.

Trajectory data are first generated on the basis of initial nosetip shape, assuming for the first iteration that the nosetip is completely shape stable. These trajectory data are input to a transient, axisymmetric (two-dimensional), finite-difference thermal code [6] which computes the shape-coupled thermal environment and resulting instantaneous nosetip shape and indepth temperature field. If shape changes are sufficient to significantly alter drag, a second iteration on trajectory is performed. Final converged sets of instantaneous shapes, pressures, loads, and temperature fields at discrete altitudes are stored on tape and automatically input into an axisymmetric finite-element structures code [10] that calculates stress, strain, and deformation data of interest. Graphical displays of these data are used to assess design acceptability based on pre-established failure strain criteria for graphite.

The above iterative design sequence generally entails the evolution of a compromised design followed by sensitivity studies aimed at assessing probable bounds on shape stability and structural survival.

## Thermal Response Model

Indeath thermal response of axisymmetric nosetips at zero angle of attack is calculated by a transient, finite-difference scheme [6] using fixed rectangular grids with nonuniform spacing (Fig. 5). The shell conduction network provides for various materials  $M_N$  with interface contact resistance, and features fine grid spacing in regions of anticipated high thermal strain. The plug model assumes an adiabatic shank surface, consistent with the use of a surrounding heatshield of low conductivity.

The major uncertainties in the thermal response model concern input data on boundary-layer transition and graphite ablation. Figure 6 shows the general range of transition-onset data deduced from ground and unclassified flight tests. Flight transition data are derived from nosetip sidewall sensors (high  $Ma$ ), and uncertainty exists in applying these data to the forward nosetip region (low  $Ma$ ) where the major shape changes occur. Nosetip strain sensitivity studies have been made with various limiting transition criteria, ranging from constant  $Re_{0,c}$  values to the Mach-number dependent curves shown in Fig. 6. Once transition occurs, a variable transitional heating function is used to

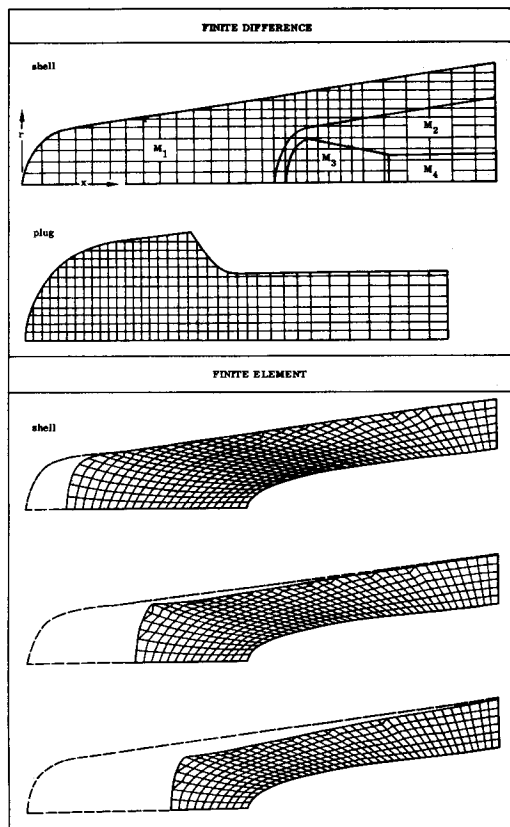


FIG. 5. THERMAL AND STRUCTURAL NUMERICAL ANALYSIS GRIDS.

span the pure smooth-wall laminar and fully developed smooth-wall turbulent heating regimes.

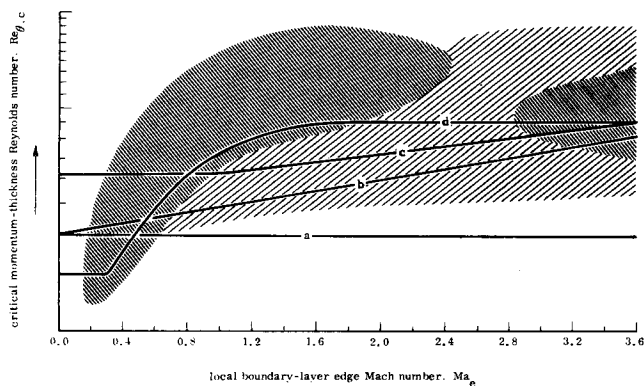


FIG. 6. BOUNDARY-LAYER TRANSITION-ONSET MODELS.

Figure 7 gives a comparison of the mass-transfer parameter  $B_w$  for carbon sublimation (ratio of carbon mass flux to blowing heat-transfer coefficient) and resulting surface temperature  $t_w$  and ablation rate  $\dot{s}$  from a number of contemporary theories [7,11,12,13]. While each of the equilibrium models depends essentially on the same boundary-layer mass and energy transfer assumptions, differences exist in the reaction system adopted and in the thermodynamic properties assigned to the various ablation species. The kinetic rate-controlled model [11] assumes that reaction kinetics are not infinitely fast as assumed in the equilibrium models. This theory is therefore more appealing, except for its uncertainties in vapor-

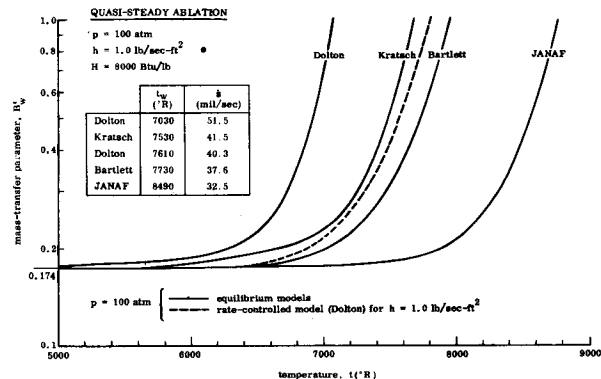


FIG. 7. CHEMICAL EROSION MODELS FOR GRAPHITE SUBLIMATION.

ization coefficients [14]. None of the models is conclusively substantiated by experimental data, one difficulty being that mechanical erosion cannot be separated from chemical ablation at the high pressures of interest.

At high pressures measured graphite recession rates begin to depart from pure chemical erosion due to mechanical loss effects. Figure 8 shows two mechanical erosion models used in the nosetip design sensitivity studies. According to theory I, chem-

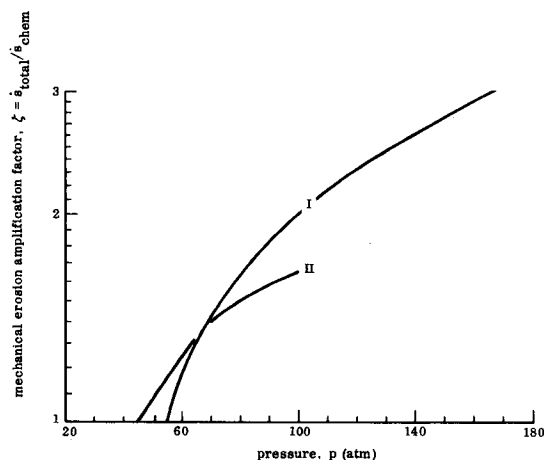


FIG. 8. MECHANICAL EROSION MODELS FOR GRAPHITE.

ical erosion alone controls surface recession at all locations along the nosetip where  $p \leq 55$  atm, while at locations where  $p > 55$  atm mechanical erosion combines with chemical ablation to amplify total surface recession. At  $\zeta = 3$ , for example, the mechanical erosion rate is twice the chemical rate.

Figure 9 gives representative thermal and mechanical properties of graphite used in design calculations of thermal and structural response. Except for specific heat  $c$ , the properties of thermal conductivity  $k$ , thermal expansion  $\beta$ , elastic modulus  $E$ , and Poisson ratio  $\nu$  are directionally dependent owing to the oriented grain structure of graphite. The bounding curves give uniaxial properties that have been measured along the orthogonal directions across and with grain.

Figure 10 illustrates how a graphite nosetip dissipates and accommodates reentry heating. The second peak in curve 1 is caused by transition and subsequent decrease in stagnation-point radius of curvature. Curve 2 shows the strong hot-wall effect

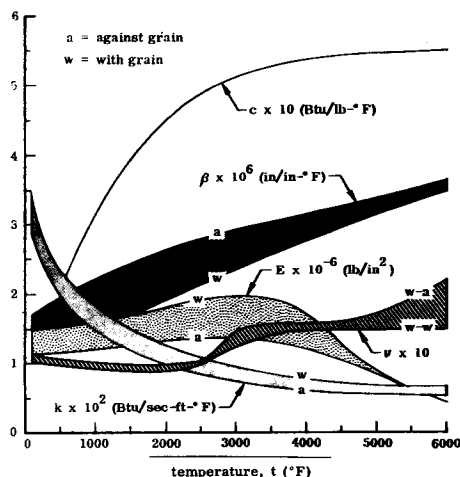


FIG. 9. THERMAL AND MECHANICAL PROPERTIES OF GRAPHITE.

with high graphite surface temperatures, and curve 3 gives the effect of mass transfer of carbon gas species. Curve 4 shows energy absorption by surface oxidation and sublimation. The remaining heat rate

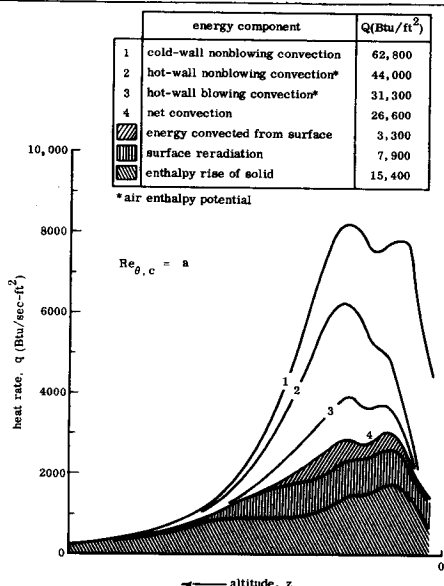


FIG. 10. ENERGY PARTITION AT STAGNATION POINT OF GRAPHITE NOSETIP.

(down by about 60% near peak heating of its original cold-wall nonblowing value) is partitioned into three energy accommodation components as shown by the crosshatched regions, namely net energy convected at the surface (energy convected away by blowing minus solid enthalpy convected to wall), energy emitted by surface radiation, and enthalpy rise of the surviving solid. Thus, the dominant thermal-protection effects with a graphite nosetip are high surface temperature (hot-wall and reradiation effects), mass-transfer cooling (blowing), and high thermal capacity.

A correlation of total impact recession  $s$  at the stagnation point of graphite nosetips, normalized by original nosetip radius  $R_N$ , is given in Fig. 11. The independent variable is a dimensionless reentry parameter  $\lambda$  formed by considering that  $s$  is proportional to the total laminar convective heat load. In subsequent diagrams, comparative values of  $\lambda$  are given

to indicate relative severity of the environment as determined by  $V_E$ ,  $\delta_E$ ,  $\beta$  and  $R_N$ . The large number of calculated points shown in Fig. 11 (scatter caused by varying transition and erosion models)

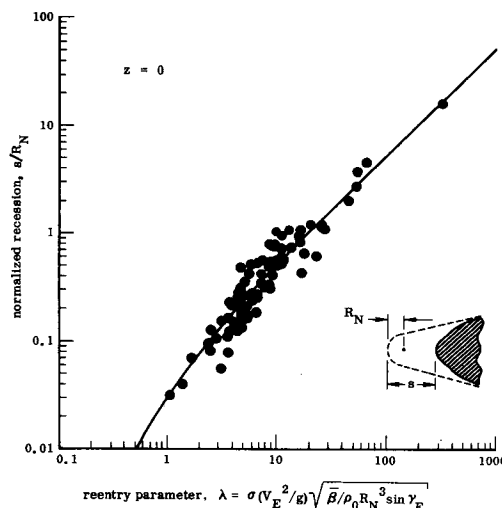


FIG. 11. NOSETIP STAGNATION-POINT RECESSION CORRELATIONS.

suggests the number of separate nosetip designs and associated sensitivity studies conducted by IMSC.

### Structural Response Model

Thermal strains in the graphite nosetip at zero angle of attack are calculated by an elastic/plastic, axisymmetric, finite-element scheme [10] which automatically generates a new grid for the ablated shape at each successive altitude to be investigated (Fig. 5). The code includes three different quadrilateral linear strain elements, an eight degree-of-freedom isoparametric element and two modified isoparametric elements for improved bending response. Inputs to the code include orthotropic material properties (Fig. 9). Solutions of the linear simultaneous network equations are obtained by Gaussian elimination techniques for determination of stresses, strains, and displacements throughout the finite-element model. An iterative scheme is used to account for differences between tensile and compressive properties. Owing to inadequate material characterization, only the linear elastic portion of the structures code has been used here.

### Thermal/Structural Performance Predictions

**Plug design.** Figure 12 gives the stagnation-point heating history and combined nosetip/heatshield ablation contours at impact for a blunt graphite plug nosetip exposed to initial reentry conditions represented by the four corners of a rectangular  $V$ - $\delta$  map. The ablation contours were obtained by a coupled calculation using a quasi-steady shape-change code [15] modified for two ablative materials. At low initial velocity  $V_E$  the total stagnation-point and heatshield recessions are nearly independent of the reentry angles considered and only the shoulder region of the plug feels the effect of longer reentry time at low  $\delta_E$  (trajectory ①). At high velocity both the plug and heatshield recessions are sensitive to  $\delta_E$ , the more severe environment at large  $\delta_E$  dominating the shape change (trajectory ④).

Figure 13 shows instantaneous shape and indepth isotherms in a blunt graphite plug nosetip for upper-

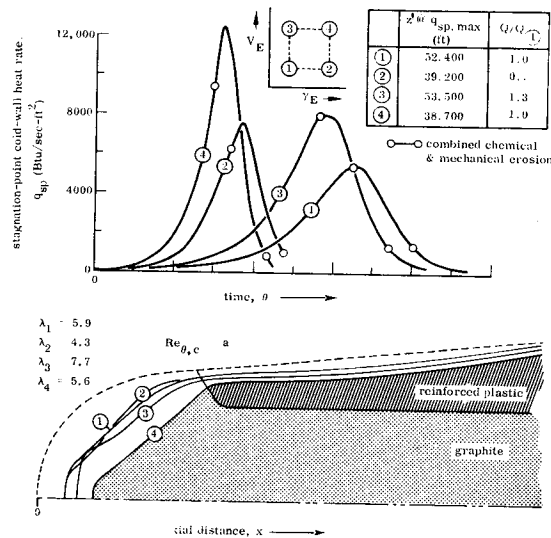


FIG. 12. EFFECT OF TRAJECTORY ON IMPACT SHAPE OF PLUG/HEATSHIELD COMBINATION.

right (steep reentry) and upper-left (shallow reentry) trajectories and early transition. Although stagnation-point recession in the steep reentry case lags that for shallow reentry at medium altitudes, it catches up near  $z = 25$  kft and exceeds shallow reentry at impact. Plug shank temperatures are

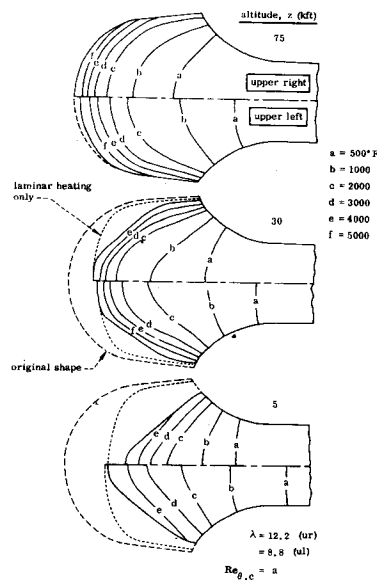


FIG. 13. EFFECT OF TRAJECTORY ON SHAPE AND TEMPERATURE RESPONSE OF A GRAPHITE PLUG NOSETIP.

higher for upper left due to prolonged conduction time, the rear-shank temperatures near impact being in the neighborhood of  $500^\circ\text{F}$  and the forward-shank temperatures approaching  $5000^\circ\text{F}$  before impact.

Figure 14 gives isotherms and corresponding isostrains in the  $r$ - $x$  (pitch) plane of a graphite plug nosetip for variable (delayed) transition. Maximum tensile strains develop along the plug centerline, their peaks occurring near impact.

Figure 15(a) shows instantaneous contours and location of boundary-layer transition (BLT) on a graphite plug nosetip as effected by transition criteria (Fig. 6) and by combination of chemical

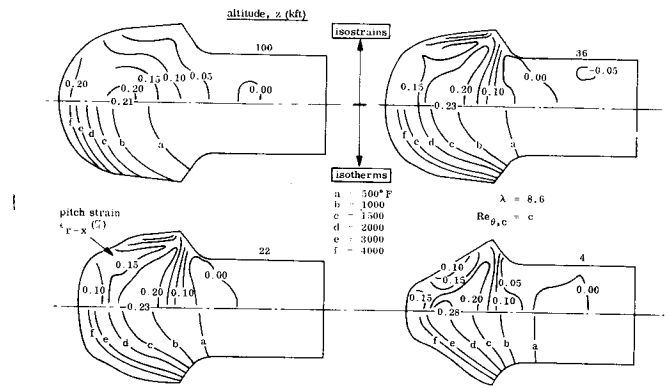


FIG. 14. ISOTHERMS AND ISOSTRAINS IN A GRAPHITE PLUG NOSETIP.

erosion model (Dolton or Bartlett, Fig. 7) and mechanical erosion model (I or II, Fig. 8). Figure

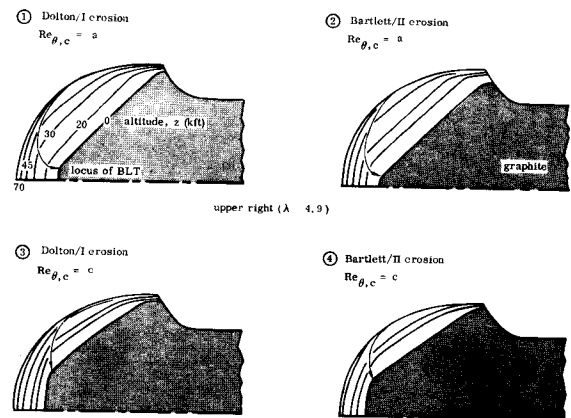


FIG. 15. EFFECT OF CHEMICAL/MECHANICAL EROSION MODEL AND TRANSITION ON GRAPHITE PLUG NOSETIP: (a) SHAPE HISTORY.

15(b) gives additional comparisons for these modeling assumptions. Effective nosetip radius of curvature effects aerodynamics through nosetip pressure drag. Progressive stagnation-point blunting develops if no transition occurs. Blunting in the stagnation region

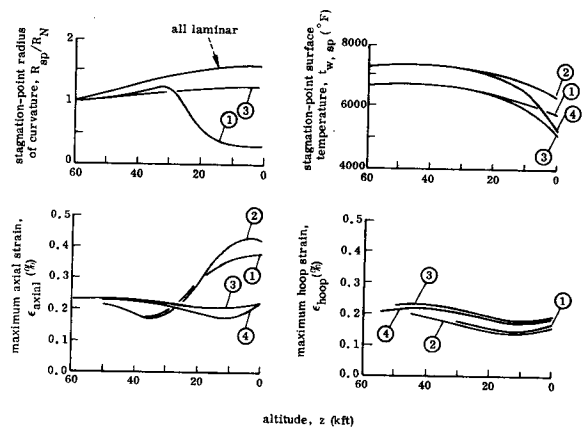


FIG. 15. CONCLUDED: (b) SURFACE CHARACTERISTICS AND THERMAL STRAINS.

also develops for variable transition, case ③, while early transition, case ①, causes abrupt sharpening at low altitudes. Stagnation-point surface temperatures depend on ablation model at high altitudes and on transition model at low altitudes. The maximum axial and hoop strains in the plug are

relatively insensitive to ablation model and depend mostly on transition criterion.

The effect of transition on the history of maximum strain in a graphite plug nosetip is illustrated in Fig. 16. Delayed transition markedly reduces maximum axial strain and causes a slight increase in maximum hoop strain; this changes the potential fail-

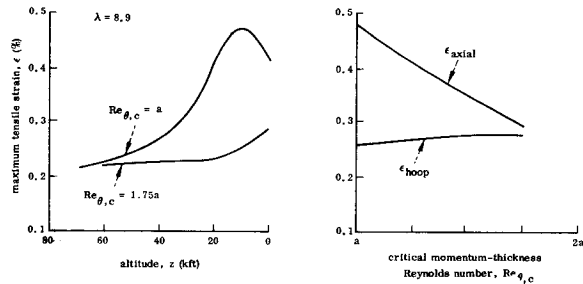


FIG. 16. EFFECT OF TRANSITION ON HISTORY OF MAXIMUM STRAINS IN A GRAPHITE PLUG NOSETIP.

ure mode from radial cracks to axial cracks (Fig. 2).

Figure 17 gives the history of maximum axial and hoop strains in a graphite plug nosetip for reentry

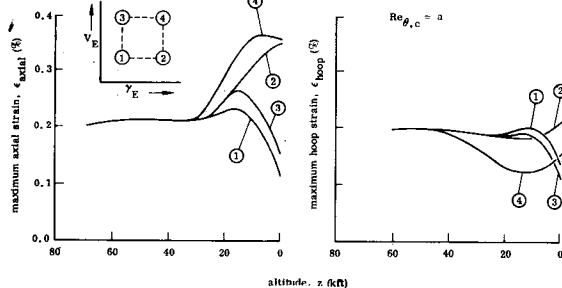


FIG. 17. EFFECT OF TRAJECTORY ON HISTORY OF MAXIMUM STRAINS IN A GRAPHITE PLUG NOSETIP.

at the four corners of a  $V$ - $\gamma$  map. Trajectory effects on thermal strain are delayed to low altitudes in the plug configuration. The peaks in axial strain are higher when reentry velocity and/or angle is increased. Both the maximum axial and maximum hoop strains fall off at low altitudes for shallow reentry, owing to the plug becoming more isothermal during the longer heating periods associated with low  $\gamma_E$ .

Concerning the geometry of plug nosetips, while the diameter and length of shank are important in

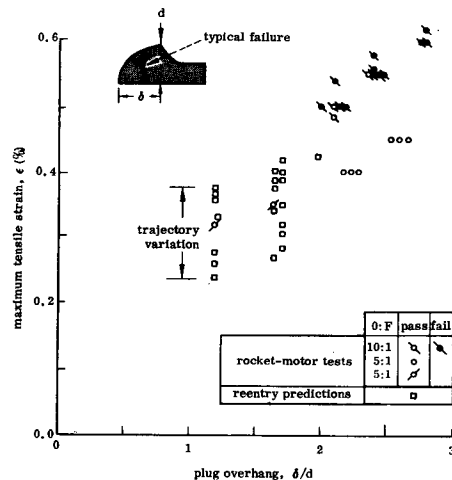


FIG. 18. THERMOSTRUCTURAL GROUND TESTS ON GRAPHITE PLUG NOSETIPS.

satisfying certain loading conditions and forward heatshield performance, maximum thermal strains are sensitive to the plug overhang  $\delta$  and overhang radius  $d$ . Figure 18 gives calculated data on maximum tensile strain in graphite plug nosetips for design reentry cases and for actual exposure in rocket-motor ground testing. The rocket-motor tests produced thermal strain failures at  $\delta/d \approx 2$  and where the test environment (oxygen-to-fuel ratio) was adjusted to produce theoretical strains at and above  $\epsilon = 0.50$  %. At high  $\delta/d$  the axial strain becomes more critical and radial cracks are expected. This behavior was confirmed in the ground tests.

**Shell design.** Figure 19 shows isotherms and corresponding isostrains in a blunt graphite shell nosetip

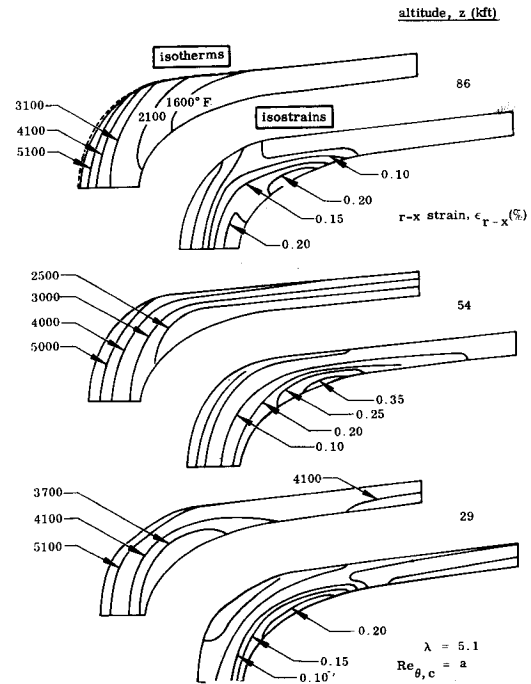


FIG. 19. ISOTHERMS AND ISOSTRAINS IN A GRAPHITE SHELL NOSETIP.

having an elliptical inner surface contour. This shape produces lower strains than a hemisphere/cone surface. Maximum strains develop in the vicinity of the ellipse/cone tangency point. Peak strains occur near 54 kft and decrease at lower altitudes where temperature gradients diminish and the shell becomes more nearly isothermal.

Figure 20 presents the results of a linear elastic solution for the biaxial strains along the inner surface of a graphite shell nosetip. The arrows on each curve indicate the direction from the stagnation point proceeding towards the aft end of the nosetip. Maximum strains occur in the across-grain direction at 49 kft. This figure illustrates that a nearly one-to-one ratio of biaxial strains develop along the inner surface. These data are to be compared with a maximum strain failure criterion to establish nosetip survival. Figure 21 gives an example criterion based on measured biaxial strain failure on a limited number of graphite billets [16].

The effect of boundary-layer transition on maximum thermal strains in a sharp graphite shell nosetip is illustrated in Fig. 22. As the transition criterion is changed from a constant (early)  $Re_{\theta,c}$

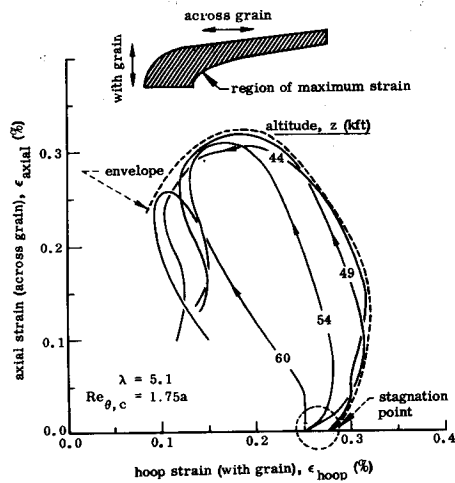


FIG. 20. BIAxIAL STRAINS IN A GRAPHITE SHELL NOSETIP.

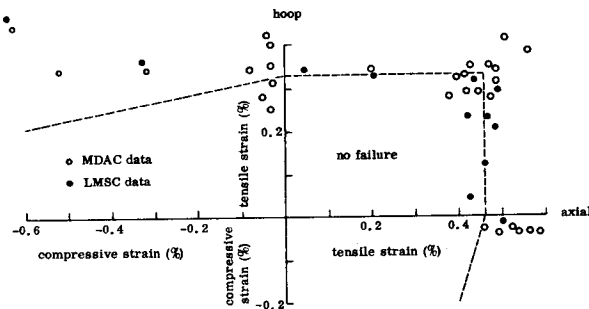


FIG. 21. BIAxIAL STRAIN ALLOWABLES FOR GRAPHITE.

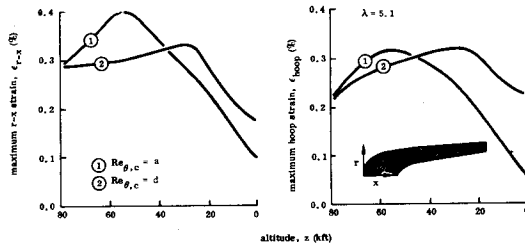


FIG. 22. EFFECT OF TRANSITION ON HISTORY OF MAXIMUM STRAINS IN A GRAPHITE SHELL NOSETIP.

to a variable (delayed)  $Re_{\theta,c}$ , the peak strains move from the tangency region toward the stagnation point. The peak in r-x strain history is delayed and decreased, while in the hoop direction this peak is delayed and increased. Thus, transition can effect both failure mode and fracture direction.

Figure 23 shows the history of maximum tensile strain (r-x or hoop near the inner stagnation point) in a blunt graphite shell nosetip as a function of initial reentry conditions. Increasing either or both the reentry velocity and angle generally increases the peak in maximum strain. The variation in these peak tensile strains across the V- $\gamma$  map is shown in Fig. 24 for two constant transition criteria. Later transition (larger  $Re_{\theta,c}$ ) reduces the peak strain, the relief being larger in the middle of the V- $\gamma$  map than at either the upper-left or upper-right corners.

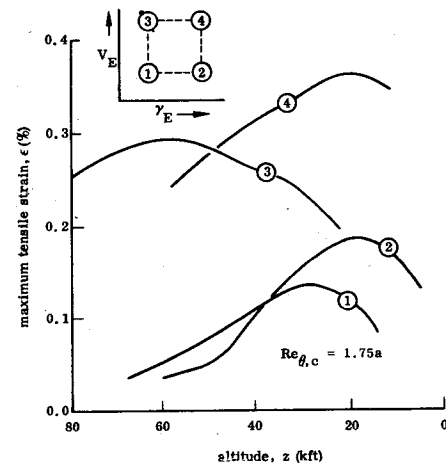


FIG. 23. EFFECT OF TRAJECTORY ON HISTORY OF MAXIMUM STRAINS IN A GRAPHITE SHELL NOSETIP.

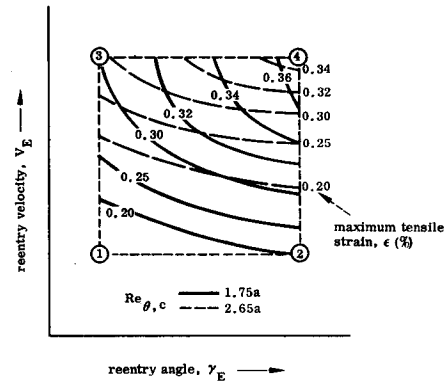


FIG. 24. EFFECT OF TRANSITION ON V- $\gamma$  ISOstrain MAP FOR A GRAPHITE SHELL NOSETIP.

A summary of more extensive calculations for shell and plug nosetips is given in the isostrain V- $\gamma$  plots of Fig. 25. Although an increase in either  $V_E$  or  $\gamma_E$  generally increases maximum tensile thermal strains, there are conditions where the relieving

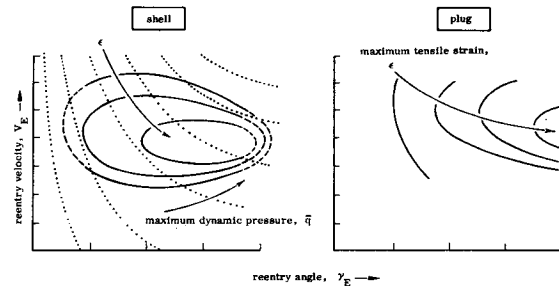


FIG. 25. V- $\gamma$  ISOstrain MAPS FOR GRAPHITE SHELL AND PLUG NOSETIPS.

effects of erosion and external air pressure (dotted curves) overtakes the increasing thermal strain. The net result is that strains begin to diminish rather than increase indefinitely. This causes the peak thermal strain to occur at some point in the central region of the V- $\gamma$  map and the isostrain lines to become elliptical in shape. Thus, at more severe reentry conditions, thermal strain can become less critical and other nosetip failure modes can dominate

## Special Design Problems

**Heatshield response with plug nosetip.** The choice of plug overhang entails a design compromise. While thermal strains are minimized by small plug overhang  $\delta$  (Fig. 18), increasing  $\delta$  eliminates a forward-facing plug/heatshield interface, minimizes plug/heatshield ablation mismatch, and minimizes downstream heatshield ablation (Fig. 26). For larger overhangs  $\delta/R_N$ , the graphite plug shields the afterbody and maximum heatshield ablation occurs downstream of the material discontinuity ("corkbottle" shape change).

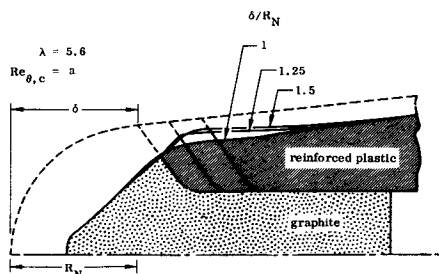


FIG. 26. EFFECT OF PLUG OVERHANG  $\delta$  ON HEATSHIELD ABLATION.

Figure 27 shows progressive ablation and in-depth decomposition of a plastic heatshield used with a blunt graphite plug nosetip. Also

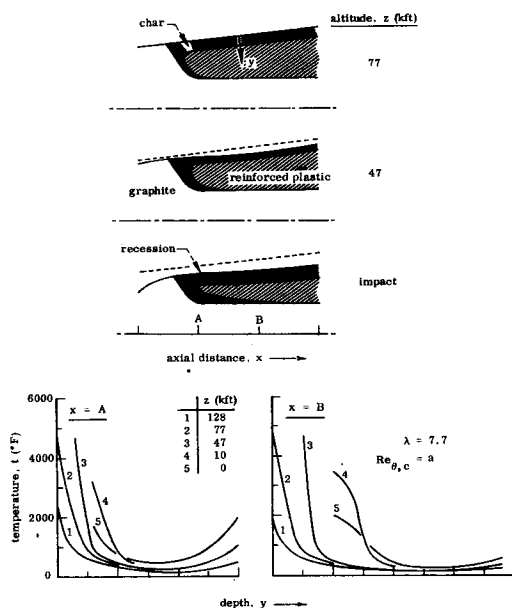


FIG. 27. HEATSHIELD DECOMPOSITION WITH GRAPHITE PLUG NOSETIP.

shown are instantaneous temperature profiles through the surviving heatshield at two axial stations. High interface temperatures can cause local heatshield decomposition and aggravated ablation effects near the material discontinuity (Fig. 2). In addition, backface heatshield charring occurs due to contact with the hot plug shank (Fig's. 13 and 14). Phenolic decomposition gases must be vented through this developing char layer. At low altitudes the outer heatshield begins to cool while the temperature of its inner surface continues to rise. This can cause aggravated thermal stresses and local failures in the degraded and pressurized forward heatshield region.

**Thermal growth of shell nosetip.** The need to main-

tain lateral support while the nosetip is continuously expanding represents one of the more difficult requirements to satisfy in shell nosetip designs (Fig. 2). Relative axial growth at the inner stagnation point and radial growth at the sidewall attachment region are shown in Fig. 28, along with required axial movement of the supporting wedge. This seg-

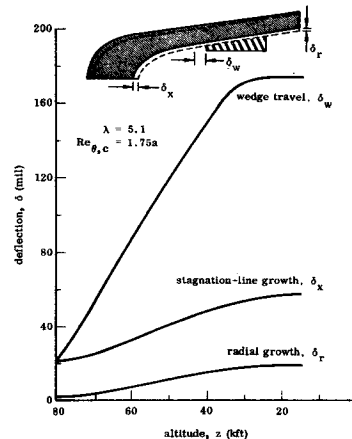


FIG. 28. THERMAL GROWTH OF A GRAPHITE SHELL NOSETIP.

mented wedge is spring loaded in the axial direction to insure that it slides forward during nosetip assembly and during heatup, thereby providing continuous lateral support throughout reentry.

**Flow through shell nosetip.** As a result of the finite open porosity of graphite, a small fraction of high-pressure external flow can leak into the graphite shell and penetrate the shell cavity (Fig. 2). This can cause internal oxidation of the porous shell (degrading mechanical properties), superimpose pressure strains on existing thermal strains in the shell, and pressurize the cavity (induce aggravated attachment loads). Figure 29 shows example results from a transient compressible solution [17] for air ingestion during reentry. The pressure field at 32 kft is shown for a shell of uniform permeability whose backface is either permeable or impermeable. Also shown for the permeable case is cavity pressure and accumulated air mass in the cavity (accounting for leakage back out through the low-pressure sidewall) down to impact. The mass flow rates in the shell itself are too small to effect the matrix temperature distribution, and as such shell thermal strains are perturbed only by pore pressure.

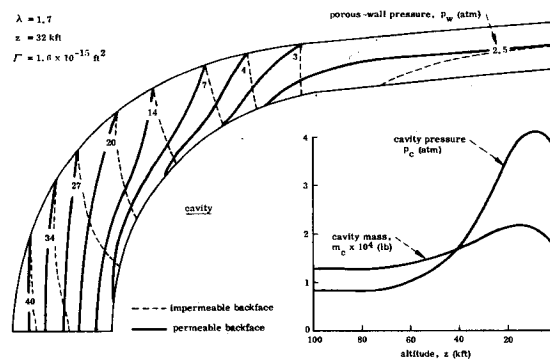


FIG. 29. FLOW IN A GRAPHITE SHELL NOSETIP.

### Conclusions

Reentry performance of shell and plug graphite nosetips has been compared. A shell nosetip is mechanically intricate, being sensitive to shape change, thermal growth, and lateral loads. The plug design is mechanically simpler but its short overhang causes it to strongly influence forward heatshield performance. Both designs are susceptible to large-scale fracture from excessive thermal strains.

Uncertainties in thermal response include transition and ablation, the former having the largest effect on thermal strain. Uncertainties in structural response center on inadequate material characterization relating to biaxial strain behavior, non-linear effects, and microstructure flaws.

The general level of thermal strains are similar in shell and plug nosetips, peak values occurring at higher altitudes for shell designs. The counteracting effects of tensile strain and compressive air loads causes peak strains to develop at V-X reentry conditions intermediate between upper left and upper right. The majority of predicted peak flight strains lie between 0.3 and 0.4% — near the maximum strain allowables for average graphite billet material.

### Acknowledgements

The analyses reported here were supported in part under Air Force ABRES contracts, Space and Missile Systems Organization, Norton AFB, California: RVT0-2B, FO4701-68-C-0157 (1968-1971); NDAT, FO4701-69-C-0237 (1969-1971); and BPD, FO4701-C-0011 (1970-1972). The authors are pleased to credit L. Brick and R. E. Wilson as LMSC managers of the supporting contracts, plus L. W. Gallagher, Y. J. Kaku and W. K. Skeehan as contributors to the design work described here.

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